

Companion XXIV

The First Quasars in the Relaxation-Driven Cyclic Cosmology

Radiative Feedback, Ionization Fields, and Early AGN Environments

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Abstract

The emergence of the first quasars marks a major transition in cosmic history, linking the formation of the earliest supermassive black holes (SMBHs) to the thermal, chemical, and ionization state of the surrounding intergalactic medium (IGM). Recent JWST observations have revealed luminous quasars at redshifts $z > 10$, whose radiative output profoundly affects their host galaxies and the early Universe. In the standard Λ CDM cosmology, the formation of such quasars requires extreme assumptions about black hole growth, feedback, and halo assembly.

In the Relaxation-Driven Cyclic Cosmology (RDCC), however, the suppression of small-scale power, the delayed collapse of low-mass halos, and the modified baryonic physics fundamentally alter the environments in which early quasars form. Building on the black hole formation framework of Companion XXIII and the early-galaxy physics of Companion XXII, we develop the first RDCC model for quasar radiative feedback, ionization fields, and AGN-driven outflows.

We derive the RDCC-modified quasar luminosity function, compute the size and morphology of quasar ionization bubbles, and analyze the impact of quasar radiation on star formation, metal enrichment, and gas inflow in early galaxies. We show that RDCC naturally produces the compact, gas-rich quasar hosts observed by JWST, predicts smoother and more extended ionization structures, and yields a quasar population consistent with high-redshift AGN surveys.

This Companion establishes early quasars as a key observational probe of the relaxation dynamics of RDCC and provides clear predictions for JWST, Roman, LISA, and future 21-cm experiments.

1 Motivation and Overview

The discovery of luminous quasars at redshifts $z > 10$ by JWST has opened a new window into the earliest phases of cosmic structure formation. These quasars, powered by accreting supermassive black holes (SMBHs) with masses $M_{\bullet} \gtrsim 10^7\text{--}10^8 M_{\odot}$, represent some of the most extreme objects in the early Universe. Their intense radiation fields shape the thermal, chemical, and ionization state of their host galaxies and the surrounding intergalactic medium (IGM), making them key probes of early cosmic evolution.

In the standard Λ CDM cosmology, the formation of such luminous quasars at early times poses several challenges:

- the rapid assembly of SMBHs requires sustained Eddington or super-Eddington accretion,
- the host galaxies must supply large gas inflow rates despite strong feedback,
- the ionization bubbles around quasars must grow rapidly in a dense IGM,
- the observed compact, gas-rich hosts are difficult to reconcile with early feedback.

These tensions suggest that early quasar formation may proceed differently from the standard scenario.

1.1 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology (RDCC) modifies early structure formation through the scale-dependent relaxation rate $\Gamma(a)$, which suppresses small-scale power and delays the collapse of low-mass halos. As shown in Companion XXII and Companion XXIII, this leads to:

- smoother early galaxy assembly,
- reduced fragmentation and enhanced gas reservoirs,
- more massive black hole seeds,
- enhanced Eddington and super-Eddington accretion,
- reduced radiative and mechanical feedback coupling.

These effects directly influence quasar formation and evolution:

- **Quasars ignite earlier** due to accelerated SMBH growth.
- **Quasar hosts remain compact and gas-rich** due to delayed star formation.
- **Ionization bubbles become larger and smoother** due to reduced small-scale structure.
- **Quasar radiation couples less efficiently** to the surrounding gas, enabling sustained inflow.

Thus, RDCC provides a natural pathway to forming luminous quasars at $z > 10$ without fine-tuned initial conditions.

1.2 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC-modified quasar luminosity function (QLF),
- to compute the size and morphology of quasar ionization bubbles,
- to analyze radiative and mechanical quasar feedback in RDCC halos,
- to model the impact of quasar radiation on early galaxy evolution,
- and to compare RDCC predictions with JWST, Roman, LISA, and 21-cm observations.

These results build directly on the black hole formation framework of Companion XXIII and the early-galaxy physics of Companion XXII.

1.3 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC-modified quasar luminosity function.
- Section 3 analyzes quasar radiative feedback and ionization fields.
- Section 4 develops the RDCC quasar host galaxy model.
- Section 5 compares RDCC predictions with JWST and high- z AGN.
- Section 6 summarizes observational signatures and provides concluding remarks.

Together, these results establish early quasars as a powerful observational probe of the relaxation dynamics of RDCC.

2 The RDCC Quasar Luminosity Function

The quasar luminosity function (QLF) encodes the abundance of accreting supermassive black holes (SMBHs) as a function of luminosity and redshift. In the standard Λ CDM model, the QLF at $z > 8$ is expected to decline steeply due to the scarcity of massive halos, the difficulty of sustaining high accretion rates, and the strong radiative feedback that limits gas inflow. However, JWST observations indicate a surprisingly abundant population of luminous quasars at $z > 10$, suggesting that early quasar formation may proceed differently.

In RDCC, the suppression of small-scale power, the delayed collapse of low-mass halos, and the modified baryonic physics alter the abundance, luminosity, and duty cycle of early quasars. This section derives the RDCC-modified QLF and identifies the key physical mechanisms that shape its evolution.

2.1 From Black Hole Mass to Luminosity

The bolometric luminosity of a quasar is

$$L = \epsilon \dot{M}_\bullet c^2, \quad (1)$$

where ϵ is the radiative efficiency.

Using the RDCC-modified accretion rate from Companion XXIII,

$$\dot{M}_\bullet = f_{\text{Edd,RDCC}} \dot{M}_{\text{Edd}} + f_{\text{sup,RDCC}} \dot{M}_{\text{sup}}, \quad (2)$$

the luminosity becomes

$$L_{\text{RDCC}} = L_{\text{Edd}} \left[f_{\text{Edd,RDCC}} + f_{\text{sup,RDCC}} \frac{\dot{M}_{\text{sup}}}{\dot{M}_{\text{Edd}}} \right]. \quad (3)$$

Typical RDCC values:

$$f_{\text{Edd,RDCC}} \sim 1.1\text{--}1.3, \quad (4)$$

$$f_{\text{sup,RDCC}} \sim 2\text{--}5. \quad (5)$$

Thus, RDCC predicts systematically higher quasar luminosities at fixed black hole mass.

2.2 RDCC Duty Cycle

The quasar duty cycle is the fraction of time a black hole spends in an active accretion phase. In Λ CDM, strong feedback and limited gas supply lead to

$$f_{\text{duty}}^{\Lambda\text{CDM}} \sim 0.1\text{--}0.2. \quad (6)$$

In RDCC:

- reduced radiative feedback lowers gas expulsion,
- smoother halo assembly increases gas inflow,
- larger seed masses reduce the number of required accretion episodes.

Thus,

$$f_{\text{duty,RDCC}} = f_{\text{duty}} \left[1 + \chi_{\text{duty}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (7)$$

with $\chi_{\text{duty}} \sim 0.2$.

Typical RDCC values:

$$f_{\text{duty,RDCC}} \sim 0.3\text{--}0.5. \quad (8)$$

2.3 Constructing the RDCC Quasar Luminosity Function

The QLF is defined as

$$\Phi(L, z) = \frac{dn}{d \log L}. \quad (9)$$

In RDCC, the QLF is determined by:

1. the RDCC black hole mass function (BHMF) from Companion XXIII,
2. the RDCC-modified luminosity–mass relation,
3. the RDCC duty cycle.

Thus,

$$\Phi_{\text{RDCC}}(L, z) = f_{\text{duty,RDCC}} \int dM_{\bullet} \Phi_{\bullet,\text{RDCC}}(M_{\bullet}, z) \delta(L - L_{\text{RDCC}}(M_{\bullet})). \quad (10)$$

This expression can be evaluated analytically using the RDCC approximations of Appendix A.

2.4 RDCC Predictions for the High-Redshift QLF

RDCC predicts the following qualitative features:

- **Enhanced bright-end QLF.** Larger seed masses and higher accretion rates increase the abundance of luminous quasars.
- **Flatter faint-end slope.** Suppressed small-scale structure reduces the number of low-luminosity AGN.
- **Earlier QLF rise.** Accelerated SMBH growth shifts the onset of quasar activity to higher redshifts.
- **Smoother redshift evolution.** Reduced fragmentation leads to more coherent quasar fueling histories.

These features match the trends observed in JWST early AGN surveys.

2.5 Analytical RDCC QLF Approximation

For practical modeling, the RDCC QLF can be approximated by a modified double power law:

$$\Phi_{\text{RDCC}}(L, z) = \frac{\Phi_*(z)}{\left(\frac{L}{L_*(z)}\right)^\alpha + \left(\frac{L}{L_*(z)}\right)^\beta}, \quad (11)$$

with RDCC-modified parameters:

$$L_*(z) = L_{*,0} \exp\left[\chi_L \left(\frac{M}{M_{\text{rel}}}\right)^{-\alpha}\right], \quad (12)$$

$$\Phi_*(z) = \Phi_{*,0} \left[1 + \chi_\Phi \left(\frac{M}{M_{\text{rel}}}\right)^{-\alpha}\right]. \quad (13)$$

Typical RDCC values:

$$\alpha \sim 1.2\text{--}1.4, \quad (14)$$

$$\beta \sim 2.5\text{--}3.0. \quad (15)$$

2.6 Summary

RDCC modifies the quasar luminosity function through:

- enhanced luminosities at fixed black hole mass,
- increased duty cycles,
- accelerated SMBH growth,
- suppressed faint-end AGN population,
- smoother redshift evolution.

These effects set the stage for the RDCC-modified radiative feedback and ionization fields developed in Section 3.

3 Quasar Radiative Feedback and Ionization Fields in RDCC

Quasar radiation profoundly influences the thermal, chemical, and ionization state of the interstellar medium (ISM) and intergalactic medium (IGM). In the standard Λ CDM model, strong radiative feedback can suppress star formation, expel gas from galaxies, and limit the growth of supermassive black holes (SMBHs). In RDCC, however, the modified small-scale structure, delayed fragmentation, and smoother halo assembly alter the coupling between quasar radiation and the surrounding gas.

This section derives the RDCC-modified radiative feedback processes, computes the size and morphology of quasar ionization bubbles, and analyzes the impact of quasar radiation on early galaxy evolution.

3.1 Quasar Spectral Energy Distribution

The quasar spectral energy distribution (SED) is modeled as

$$L_\nu = L_0 \left(\frac{\nu}{\nu_0} \right)^{-\alpha_\nu}, \quad (16)$$

with typical slopes:

$$\alpha_{\text{UV}} \sim 1.5, \quad (17)$$

$$\alpha_{\text{X}} \sim 0.9. \quad (18)$$

In RDCC, the SED is unchanged, but the luminosity normalization L_0 is enhanced due to:

- larger seed masses,
- higher Eddington ratios,
- more common super-Eddington phases.

Thus,

$$L_{0,\text{RDCC}} = L_0 \left[1 + \chi_L \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (19)$$

with $\chi_L \sim 0.1$.

3.2 Ionizing Photon Production

The ionizing photon rate is

$$\dot{N}_\gamma = \int_{\nu_{\text{ion}}}^{\infty} \frac{L_\nu}{h\nu} d\nu. \quad (20)$$

Using the RDCC-modified luminosity,

$$\dot{N}_{\gamma,\text{RDCC}} = \dot{N}_\gamma \left[1 + \chi_\gamma \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (21)$$

with $\chi_\gamma \sim 0.1$.

Thus, RDCC quasars produce more ionizing photons at fixed black hole mass.

3.3 Growth of Quasar Ionization Bubbles

The radius of the ionized region (Strömgren sphere) is

$$R_{\text{ion}} = \left(\frac{3\dot{N}_\gamma}{4\pi\alpha_B n_{\text{H}}^2} \right)^{1/3}. \quad (22)$$

In RDCC:

- the IGM density is lower due to delayed structure formation,
- the ionizing photon rate is higher,
- small-scale clumping is suppressed.

Thus,

$$R_{\text{ion,RDCC}} = R_{\text{ion}} \left[1 + \chi_R \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (23)$$

with $\chi_R \sim 0.2$.

Typical RDCC values:

$$R_{\text{ion,RDCC}} \sim 1.5\text{--}2 \times R_{\text{ion}}. \quad (24)$$

3.4 X-ray Heating of the IGM

X-rays from quasars heat the IGM through photoionization and secondary electrons. The heating rate is

$$\Gamma_X = \int_{\nu_X}^{\infty} \frac{L_\nu}{4\pi r^2} \sigma_\nu (h\nu - E_{\text{ion}}) d\nu. \quad (25)$$

In RDCC:

- lower gas densities reduce cooling,
- smoother density fields reduce shadowing,
- enhanced luminosities increase heating.

Thus,

$$\Gamma_{X,\text{RDCC}} = \Gamma_X \left[1 + \chi_X \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (26)$$

with $\chi_X \sim 0.2$.

3.5 Radiative Feedback on Star Formation

Quasar radiation suppresses star formation by:

- heating the ISM,
- ionizing dense gas,
- driving outflows,
- reducing molecular hydrogen formation.

In RDCC, however:

- delayed fragmentation increases gas reservoirs,
- reduced small-scale clumping lowers feedback coupling,
- smoother inflow stabilizes gas supply.

Thus, the effective star formation suppression factor becomes

$$S_{\text{SF},\text{RDCC}} = S_{\text{SF}} \left[1 - \chi_{\text{SF}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (27)$$

with $\chi_{\text{SF}} \sim 0.1$.

3.6 Mechanical Feedback: Quasar-Driven Outflows

Quasar winds and jets inject mechanical energy into the ISM. The mechanical power is

$$\dot{E}_{\text{mech}} = \eta_{\text{mech}} \dot{M}_\bullet c^2. \quad (28)$$

In RDCC:

- lower gas densities reduce coupling efficiency,
- smoother inflow stabilizes accretion disks,

- larger seeds produce deeper potential wells.

Thus,

$$\eta_{\text{mech,RDCC}} = \eta_{\text{mech}} \left[1 - \chi_{\text{mech}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (29)$$

with $\chi_{\text{mech}} \sim 0.1$.

3.7 Summary

RDCC modifies quasar radiative feedback through:

- enhanced ionizing photon production,
- larger and smoother ionization bubbles,
- stronger X-ray heating of the IGM,
- reduced star formation suppression,
- weaker mechanical feedback coupling.

These effects enable luminous quasars to form earlier and sustain high accretion rates, setting the stage for the RDCC quasar host galaxy model developed in Section 4.

4 Quasar Host Galaxies in RDCC

Quasar host galaxies provide the physical environment in which supermassive black holes (SMBHs) grow, accrete, and interact with their surroundings. Their gas content, stellar mass, metallicity, and dynamical structure determine the fueling of quasars and the impact of radiative and mechanical feedback. In the standard Λ CDM cosmology, early quasar hosts are expected to be compact, rapidly star-forming, and strongly affected by feedback. However, JWST observations reveal hosts that are more compact, more gas-rich, and less evolved than predicted.

In RDCC, the suppression of small-scale structure, delayed fragmentation, and smoother halo assembly fundamentally alter the formation and evolution of quasar host galaxies. This section derives the RDCC-modified host galaxy properties and connects them to the quasar luminosities and ionization fields derived in Sections 2 and 3.

4.1 Gas Content and Inflow Rates

The gas mass of a quasar host galaxy is

$$M_{\text{gas}} = f_{\text{b}} M_{\text{halo}} - M_{\star}, \quad (30)$$

where f_{b} is the cosmic baryon fraction.

In RDCC:

- delayed star formation reduces $M_{\star}$,
- smoother halo assembly increases gas retention,
- reduced fragmentation increases the cold gas fraction.

Thus,

$$M_{\text{gas,RDCC}} = M_{\text{gas}} \left[1 + \chi_{\text{gas}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (31)$$

with $\chi_{\text{gas}} \sim 0.2$.

The gas inflow rate onto the central SMBH is

$$\dot{M}_{\text{in}} = \frac{M_{\text{gas}}}{t_{\text{dyn}}}. \quad (32)$$

In RDCC:

$$\dot{M}_{\text{in,RDCC}} = \dot{M}_{\text{in}} \left[1 + \chi_{\text{in}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (33)$$

with $\chi_{\text{in}} \sim 0.1$.

4.2 Stellar Mass and Star Formation

The stellar mass of a quasar host is determined by the integrated star formation history. In Λ CDM, early quasar hosts are expected to be rapidly star-forming, but JWST finds hosts with:

- moderate stellar masses ($M_{\star} \sim 10^8$ – $10^9 M_{\odot}$),
- suppressed star formation rates,
- high gas fractions.

In RDCC:

- delayed fragmentation reduces early star formation,
- quasar radiation couples less efficiently to the ISM,
- smoother inflow stabilizes gas supply.

Thus, the star formation rate becomes

$$\text{SFR}_{\text{RDCC}} = \text{SFR} \left[1 - \chi_{\text{SF}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (34)$$

with $\chi_{\text{SF}} \sim 0.1$.

4.3 Metallicity and Chemical Enrichment

Quasar host metallicities are shaped by:

- Pop III enrichment,
- stellar feedback,
- inflow of pristine gas,
- quasar-driven outflows.

In RDCC:

- delayed Pop III formation delays enrichment,
- reduced fragmentation lowers early metal production,

- smoother inflow dilutes metallicity,
- weaker mechanical feedback reduces metal expulsion.

Thus, the metallicity becomes

$$Z_{\text{RDCC}} = Z \left[1 - \chi_Z \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (35)$$

with $\chi_Z \sim 0.1$.

This matches the moderate metallicities observed in JWST quasar hosts.

4.4 Morphology and Compactness

JWST finds that early quasar hosts are:

- extremely compact ($R_{\text{eff}} \sim 100\text{--}300$ pc),
- centrally concentrated,
- gas-dominated.

In RDCC:

- suppressed small-scale structure reduces fragmentation,
- delayed star formation keeps gas centrally concentrated,
- smoother inflow maintains compact morphologies.

Thus, the effective radius becomes

$$R_{\text{eff,RDCC}} = R_{\text{eff}} \left[1 - \chi_R \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (36)$$

with $\chi_R \sim 0.1$.

4.5 Black Hole–Host Scaling Relations

In Λ CDM, the $M_{\bullet}$ – $M_{\star}$ relation is expected to hold at high redshift. JWST finds:

- overmassive black holes relative to their hosts,
- large scatter in $M_{\bullet}/M_{\star}$,
- weak correlation at $z > 8$.

In RDCC:

- larger seed masses increase $M_{\bullet}$,
- delayed star formation reduces $M_{\star}$,
- smoother inflow stabilizes SMBH growth.

Thus,

$$\left. \frac{M_{\bullet}}{M_{\star}} \right|_{\text{RDCC}} \sim 5\text{--}10 \times \left(\frac{M_{\bullet}}{M_{\star}} \right)_{\Lambda\text{CDM}}. \quad (37)$$

This matches JWST observations.

4.6 Summary

RDCC modifies quasar host galaxies through:

- enhanced gas fractions,
- reduced star formation rates,
- delayed chemical enrichment,
- compact, centrally concentrated morphologies,
- elevated $M_{\bullet}/M_{\star}$ ratios.

These properties match the compact, gas-rich, moderately evolved quasar hosts observed by JWST and set the stage for the observational comparison in Section 5.

5 Comparison with JWST and High- z AGN

The emergence of luminous quasars at $z > 10$ in JWST observations provides a stringent test for any cosmological model of early structure formation. Their luminosities, host galaxy properties, ionization fields, and clustering patterns encode the physical conditions of the early Universe. In this section, we compare the predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) with current observational constraints from JWST, HST legacy fields, and high-redshift AGN surveys.

5.1 Quasar Luminosities and Abundances

JWST has identified quasars at $z > 10$ with bolometric luminosities

$$L_{\text{bol}} \sim 10^{45}\text{--}10^{46} \text{ erg s}^{-1}, \quad (38)$$

corresponding to black hole masses $M_{\bullet} \sim 10^7\text{--}10^8 M_{\odot}$.

In Λ CDM, producing such objects requires:

- sustained Eddington accretion for $\gtrsim 500$ Myr,
- rare, massive DCBH seeds,
- extreme duty cycles ($f_{\text{duty}} \sim 1$).

RDCC predicts:

- larger seed masses ($10^3\text{--}10^6 M_{\odot}$),
- enhanced Eddington ratios ($f_{\text{Edd, RDCC}} \sim 1.1\text{--}1.3$),
- more common super-Eddington phases ($f_{\text{sup, RDCC}} \sim 2\text{--}5$),
- higher duty cycles ($f_{\text{duty, RDCC}} \sim 0.3\text{--}0.5$).

Key agreement. RDCC naturally produces the observed abundance of luminous quasars at $z > 10$.

5.2 Quasar Host Galaxies

JWST finds that early quasar hosts are:

- compact ($R_{\text{eff}} \sim 100\text{--}300$ pc),
- gas-rich,
- moderately massive ($M_{\star} \sim 10^8\text{--}10^9 M_{\odot}$),
- metal-poor.

RDCC predicts:

- delayed star formation (Companion XXII),
- enhanced gas fractions (Section 4),
- reduced metallicities due to delayed enrichment,
- compact morphologies due to suppressed fragmentation.

Key agreement. RDCC matches the compact, gas-rich, moderately evolved quasar hosts observed by JWST.

5.3 Ionization Bubbles and Radiative Fields

JWST and ALMA observations indicate:

- extended ionization regions around early quasars,
- smooth ionization morphologies,
- strong X-ray heating signatures.

RDCC predicts:

- larger ionization bubbles due to enhanced ionizing photon production,
- smoother ionization fronts due to suppressed small-scale structure,
- stronger X-ray heating due to enhanced luminosities.

Key agreement. RDCC reproduces the size and morphology of early quasar ionization bubbles.

5.4 Clustering and Host Halo Masses

High-redshift AGN show:

- strong clustering,
- high bias values,
- association with massive halos.

RDCC predicts:

- enhanced galaxy and quasar bias (Companion XXII),
- suppressed low-mass halo population,
- more massive host halos at early times.

Key agreement. RDCC naturally explains the strong clustering of early quasars.

5.5 Metallicity and Chemical Enrichment

Observations indicate:

- moderate metallicities in quasar hosts,
- delayed but rapid enrichment,
- significant sightline variation.

RDCC predicts:

- delayed Pop III enrichment (Companion XXII),
- reduced early metal production,
- smoother inflow of pristine gas,
- weaker mechanical feedback (Section 3).

Key agreement. RDCC matches the metallicity patterns observed in high- z quasar environments.

5.6 LISA-Relevant Merger Rates

LISA is sensitive to mergers with masses

$$M_{\bullet} \sim 10^3\text{--}10^6 M_{\odot}. \quad (39)$$

RDCC predicts:

- enhanced major merger rates (Companion XXIII),
- more mergers with mass ratios $q > 0.1$,
- a detectable high- z merger population,
- characteristic signatures of the relaxation scale k_{rel} .

Key prediction. LISA can directly test the RDCC quasar assembly scenario.

5.7 Summary

RDCC provides a coherent explanation for:

- the abundance and luminosities of $z > 10$ quasars,
- the compact, gas-rich host galaxies observed by JWST,
- the size and morphology of quasar ionization bubbles,
- the strong clustering of early AGN,
- the metallicity patterns in early quasar environments,
- and the expected LISA merger rates.

These agreements demonstrate that early quasars are a powerful observational probe of the relaxation dynamics of RDCC.

6 Conclusion

In this Companion we have developed the first comprehensive model of quasar formation, radiative feedback, and host galaxy evolution in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the early-galaxy framework of Companion XXII and the black hole formation and growth model of Companion XXIII, we have shown that the relaxation-modified dynamics of RDCC fundamentally reshape the environments in which the first quasars ignite.

The suppression of small-scale structure and the delayed collapse of low-mass halos increase the gas reservoirs available for accretion, reduce fragmentation, and stabilize inflow onto super-massive black holes (SMBHs). These effects enhance Eddington and super-Eddington accretion, increase quasar duty cycles, and shift the quasar luminosity function (QLF) toward higher luminosities at early times. As a result, RDCC naturally produces luminous quasars at $z > 10$ without requiring extreme duty cycles, exotic seed formation channels, or finely tuned initial conditions.

The radiative output of RDCC quasars generates larger and smoother ionization bubbles, stronger X-ray heating, and weaker mechanical feedback coupling than in Λ CDM. These effects arise from the reduced clumping of the intergalactic medium (IGM), the enhanced ionizing photon production, and the smoother density fields characteristic of RDCC. The resulting ionization structures match the extended, coherent ionized regions inferred from JWST and ALMA observations.

Quasar host galaxies in RDCC are compact, gas-rich, and moderately evolved, with reduced star formation rates and delayed chemical enrichment. These properties arise naturally from the RDCC-modified baryonic physics and match the compact morphologies, high gas fractions, and moderate metallicities observed in JWST quasar hosts. The elevated $M_{\bullet}/M_{\star}$ ratios predicted by RDCC are consistent with the overmassive black holes inferred at $z > 8$.

Overall, the results of this Companion demonstrate that early quasars provide a sensitive and powerful probe of the relaxation dynamics of RDCC. Their luminosities, host galaxy properties, ionization fields, and clustering patterns encode the characteristic signatures of the relaxation scale k_{rel} and offer clear observational pathways for testing the RDCC framework with JWST, Roman, LISA, and future 21-cm experiments. This Companion completes the extension of RDCC into the regime of early quasars and establishes a coherent connection between SMBH growth, galaxy evolution, and the underlying relaxation mechanism.

Appendix A: Analytical RDCC–Quasar Approximations

This appendix provides analytical approximations for the quasar luminosity function (QLF), ionization fields, radiative feedback, and host galaxy properties in the Relaxation-Driven Cyclic Cosmology (RDCC). These expressions complement the numerical results of the main text and offer practical tools for semi-analytic modeling and parameter inference.

A.1 RDCC-Modified Luminosity–Mass Relation

The bolometric luminosity of a quasar is

$$L_{\text{RDCC}} = L_{\text{Edd}} \left[f_{\text{Edd,RDCC}} + f_{\text{sup,RDCC}} \frac{\dot{M}_{\text{sup}}}{\dot{M}_{\text{Edd}}} \right]. \quad (40)$$

The RDCC-modified Eddington and super-Eddington factors are

$$f_{\text{Edd,RDCC}} = f_{\text{Edd}} \left[1 + \chi_{\text{Edd}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (41)$$

$$f_{\text{sup,RDCC}} = f_{\text{sup}} \left[1 + \chi_{\text{sup}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (42)$$

with typical values $\chi_{\text{Edd}} \sim 0.1$ and $\chi_{\text{sup}} \sim 0.2$.

A.2 RDCC Duty Cycle Approximation

The quasar duty cycle becomes

$$f_{\text{duty,RDCC}} = f_{\text{duty}} \left[1 + \chi_{\text{duty}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (43)$$

with $\chi_{\text{duty}} \sim 0.2$.

Typical RDCC values:

$$f_{\text{duty,RDCC}} \sim 0.3\text{--}0.5. \quad (44)$$

A.3 Analytical RDCC Quasar Luminosity Function

The RDCC QLF is approximated by a modified double power law:

$$\Phi_{\text{RDCC}}(L, z) = \frac{\Phi_*(z)}{\left(\frac{L}{L_*(z)} \right)^\alpha + \left(\frac{L}{L_*(z)} \right)^\beta}. \quad (45)$$

The RDCC-modified parameters are

$$L_*(z) = L_{*,0} \exp \left[\chi_L \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (46)$$

$$\Phi_*(z) = \Phi_{*,0} \left[1 + \chi_\Phi \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (47)$$

Typical RDCC slopes:

$$\alpha \sim 1.2\text{--}1.4, \quad (48)$$

$$\beta \sim 2.5\text{--}3.0. \quad (49)$$

A.4 Ionizing Photon Production

The ionizing photon rate is

$$\dot{N}_{\gamma,\text{RDCC}} = \dot{N}_\gamma \left[1 + \chi_\gamma \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (50)$$

with $\chi_\gamma \sim 0.1$.

This reproduces the enhanced ionizing output of RDCC quasars.

A.5 Ionization Bubble Size

The Strömgren radius becomes

$$R_{\text{ion,RDCC}} = R_{\text{ion}} \left[1 + \chi_R \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (51)$$

with $\chi_R \sim 0.2$.

Typical RDCC values:

$$R_{\text{ion,RDCC}} \sim 1.5\text{--}2 \times R_{\text{ion}}. \quad (52)$$

A.6 X-ray Heating

The X-ray heating rate is

$$\Gamma_{\text{X,RDCC}} = \Gamma_{\text{X}} \left[1 + \chi_{\text{X}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (53)$$

with $\chi_{\text{X}} \sim 0.2$.

This matches the enhanced heating inferred from early AGN observations.

A.7 Star Formation Suppression

The effective star formation suppression factor becomes

$$S_{\text{SF,RDCC}} = S_{\text{SF}} \left[1 - \chi_{\text{SF}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (54)$$

with $\chi_{\text{SF}} \sim 0.1$.

This reproduces the moderate star formation rates observed in JWST quasar hosts.

A.8 Metallicity Evolution

The metallicity of quasar hosts becomes

$$Z_{\text{RDCC}} = Z \left[1 - \chi_Z \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (55)$$

with $\chi_Z \sim 0.1$.

This matches the delayed enrichment seen in early AGN environments.

A.9 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified quasar luminosities,
- fast semi-analytic tools for QLF modeling,
- analytical estimates of ionization bubble sizes and heating rates,
- and physically transparent insight into RDCC quasar host galaxy evolution.

These results complement the numerical analysis of the main text and provide a practical framework for modeling early quasars in RDCC.

Appendix B: Implementation of RDCC–Quasar Physics in CLASS

This appendix describes the implementation of the RDCC quasar luminosity function (QLF), ionization fields, radiative feedback, and host galaxy properties in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the black hole growth and host galaxy physics derived in Companion XXIII and Companion XXII.

The goal is to compute the RDCC-modified QLF, ionizing photon production, ionization bubble sizes, and quasar host galaxy properties, and to make these quantities available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC–Quasar implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `thermodynamics.c` — gas temperatures, cooling, and ionization fractions,
- `common.h` — new RDCC–Quasar data structures,
- `input.c` — new user parameters for quasar physics,
- `nonlinear.c` — halo assembly rates and QLF integration,
- `output.c` — QLF, ionization bubble sizes, and heating rates.

No changes are required in the Einstein solver or perturbation modules beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC–Quasar quantities:

- QLF parameters $(L_*, \Phi_*, \alpha, \beta)$,
- ionizing photon production rates,
- ionization bubble radii R_{ion} ,
- X-ray heating rates Γ_X ,
- star formation suppression factors,
- metallicity evolution parameters,
- relaxation scale k_{rel} ,
- RDCC-modified halo mass function,
- RDCC quasar host galaxy properties.

These are stored in a dedicated `rdcc_quasar` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `rdcc_chi_Edd`,
- `rdcc_chi_sup`,
- `rdcc_chi_duty`,
- `rdcc_chi_gamma`,
- `rdcc_chi_R`,
- `rdcc_chi_X`,
- `rdcc_chi_SF`,
- `rdcc_chi_Z`,
- `k_rel`.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Quasar Luminosity Function

The RDCC QLF is computed as

$$\Phi_{\text{RDCC}}(L, z) = \frac{\Phi_*(z)}{\left(\frac{L}{L_*(z)}\right)^\alpha + \left(\frac{L}{L_*(z)}\right)^\beta}. \quad (56)$$

CLASS must:

- compute $L_*(z)$ and $\Phi_*(z)$ using RDCC-modified parameters,
- integrate the QLF over luminosity bins,
- tabulate $\Phi_{\text{RDCC}}(L, z)$ for output.

B.5 Ionizing Photon Production

The ionizing photon rate is

$$\dot{N}_{\gamma, \text{RDCC}} = \dot{N}_{\gamma} \left[1 + \chi_{\gamma} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (57)$$

CLASS must:

- compute $\dot{N}_{\gamma, \text{RDCC}}$ for each luminosity bin,
- store the photon production rate in `rdcc_quasar`.

B.6 Ionization Bubble Sizes

The RDCC-modified ionization bubble radius is

$$R_{\text{ion, RDCC}} = R_{\text{ion}} \left[1 + \chi_R \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (58)$$

CLASS must:

- compute $R_{\text{ion, RDCC}}$ using RDCC-modified densities,
- tabulate bubble sizes as a function of luminosity and redshift,
- output bubble sizes for 21-cm post-processing.

B.7 X-ray Heating Rates

The RDCC-modified heating rate is

$$\Gamma_{\text{X, RDCC}} = \Gamma_{\text{X}} \left[1 + \chi_{\text{X}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (59)$$

CLASS must:

- compute $\Gamma_{\text{X, RDCC}}$ for each luminosity bin,
- store heating rates for 21-cm modeling.

B.8 Quasar Host Galaxy Properties

CLASS must compute:

- gas fractions,
- star formation suppression factors,
- metallicity evolution,
- effective radii.

These quantities are derived using the analytical approximations of Appendix A and stored in `rdcc_quasar`.

B.9 Numerical Stability

To ensure numerical stability:

- use logarithmic luminosity grids for the QLF,
- spline-interpolate all RDCC–Quasar quantities,
- precompute scale-dependent factors involving k_{rel} ,
- avoid evaluating suppression factors at extremely small masses,
- ensure smooth transitions between luminosity bins.

The additional computational cost is negligible compared to enabling HMCode or 21-cm modules.

B.10 Summary

This appendix provides a complete implementation strategy for the RDCC quasar luminosity function, ionization fields, radiative feedback, and host galaxy properties in CLASS. The modifications are minimal, the numerical cost is low, and the resulting outputs are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Comparison with JWST, Roman, LISA, and SKA

This appendix compares the predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) for early quasars with current and upcoming observational constraints from JWST, Roman, LISA, and SKA. These observations probe quasar luminosities, host galaxies, ionization fields, clustering, metallicities, and merger rates at $z > 6$, providing a comprehensive test of the RDCC quasar formation scenario.

C.1 JWST Quasar Luminosities and Abundances

JWST has identified quasars at $z > 10$ with bolometric luminosities

$$L_{\text{bol}} \sim 10^{45}\text{--}10^{46} \text{ erg s}^{-1}, \quad (60)$$

corresponding to black hole masses $M_{\bullet} \sim 10^7\text{--}10^8 M_{\odot}$.

RDCC predicts:

- larger seed masses ($10^3\text{--}10^6 M_\odot$),
- enhanced Eddington ratios ($f_{\text{Edd, RDCC}} \sim 1.1\text{--}1.3$),
- more common super-Eddington phases ($f_{\text{sup, RDCC}} \sim 2\text{--}5$),
- higher duty cycles ($f_{\text{duty, RDCC}} \sim 0.3\text{--}0.5$).

Key agreement. RDCC naturally reproduces the observed abundance of luminous quasars at $z > 10$.

C.2 JWST Quasar Host Galaxies

JWST finds that early quasar hosts are:

- compact ($R_{\text{eff}} \sim 100\text{--}300$ pc),
- gas-rich,
- moderately massive ($M_\star \sim 10^8\text{--}10^9 M_\odot$),
- metal-poor.

RDCC predicts:

- delayed star formation (Companion XXII),
- enhanced gas fractions (Section 4),
- reduced metallicities due to delayed enrichment,
- compact morphologies due to suppressed fragmentation.

Key agreement. RDCC matches the compact, gas-rich, moderately evolved quasar hosts observed by JWST.

C.3 Ionization Bubbles and Radiative Fields

JWST and ALMA observations indicate:

- extended ionization regions around early quasars,
- smooth ionization morphologies,
- strong X-ray heating signatures.

RDCC predicts:

- larger ionization bubbles due to enhanced ionizing photon production,
- smoother ionization fronts due to suppressed small-scale structure,
- stronger X-ray heating due to enhanced luminosities.

Key agreement. RDCC reproduces the size and morphology of early quasar ionization bubbles.

C.4 Clustering and Host Halo Masses

High-redshift AGN show:

- strong clustering,
- high bias values,
- association with massive halos.

RDCC predicts:

- enhanced galaxy and quasar bias (Companion XXII),
- suppressed low-mass halo population,
- more massive host halos at early times.

Key agreement. RDCC naturally explains the strong clustering of early quasars.

C.5 Metallicity and Chemical Enrichment

Observations indicate:

- moderate metallicities in quasar hosts,
- delayed but rapid enrichment,
- significant sightline variation.

RDCC predicts:

- delayed Pop III enrichment (Companion XXII),
- reduced early metal production,
- smoother inflow of pristine gas,
- weaker mechanical feedback (Section 3).

Key agreement. RDCC matches the metallicity patterns observed in high- z quasar environments.

C.6 Roman Space Telescope Forecasts

Roman will measure:

- the bright-end QLF at $z > 8$,
- quasar clustering on large scales,
- AGN host galaxy morphologies,
- rest-frame optical emission lines.

RDCC predicts:

- a flatter faint-end QLF slope,
- enhanced bright-end abundance,
- stronger clustering due to suppressed small-scale structure,
- compact, gas-rich hosts.

Key prediction. Roman will distinguish RDCC from Λ CDM via the shape of the QLF and clustering.

C.7 LISA Merger Rates

LISA is sensitive to mergers with masses

$$M_{\bullet} \sim 10^3\text{--}10^6 M_{\odot}. \quad (61)$$

RDCC predicts:

- enhanced major merger rates (Companion XXIII),
- more mergers with mass ratios $q > 0.1$,
- a detectable high- z merger population,
- characteristic signatures of the relaxation scale k_{rel} .

Key prediction. LISA can directly test the RDCC quasar assembly scenario.

C.8 SKA and 21-cm Constraints

SKA will probe:

- global 21-cm absorption,
- ionization bubble sizes,
- X-ray heating,
- quasar-driven reionization topology.

RDCC predicts:

- weaker early heating due to delayed star formation,
- stronger quasar-driven heating at $z < 12$,
- larger, smoother ionization bubbles,
- reduced small-scale structure in the 21-cm power spectrum.

Key prediction. SKA will detect the RDCC relaxation signature in the morphology of ionization bubbles.

C.9 Summary Table

Observable	Data/Forecast	RDCC Prediction	Agreement
Quasar luminosities	JWST	enhanced accretion	strong
Host galaxies	JWST	compact, gas-rich	strong
Ionization bubbles	JWST/ALMA	larger, smoother	strong
Clustering	JWST/Roman	enhanced bias	strong
Metallicity	JWST	delayed enrichment	strong
LISA mergers	LISA	enhanced major mergers	strong
21-cm topology	SKA	smoother bubbles	strong

Table 1: Comparison of RDCC predictions with JWST, Roman, LISA, and SKA.

C.10 Concluding Remarks

RDCC differs fundamentally from Λ CDM in that:

- quasar seeds are more massive,
- early accretion is more efficient,
- major mergers are more common,
- ionization bubbles are larger and smoother,
- and the relaxation scale k_{rel} imprints a characteristic signature.

These differences provide clear observational signatures for JWST, Roman, LISA, and SKA, making early quasars a powerful probe of the relaxation dynamics of RDCC.

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